Data Solution

Essential Concepts

Microsoft Dataverse 101:

Let's Data Model Your App

Business Applications

Charles Lakes II

Microsoft® MVP Most Valuable Professional

About Me:

- Pro-Code Developer
- Low-Code Maker
- Biz. Solution Consultant
- Middle-child

Agenda:

- No-Code Users
- Low-Code Makers
- Pro-Code Devs
- Database Engineers

Dataverse:

Microsoft Dataverse is a **secure, cloud-based data platform** designed to store and manage data used by business applications. It is part of the **Microsoft Power Platform** and integrates seamlessly with tools like **Power Apps, Power Automate, Power BI, Power Pages**, and **Dynamics 365**.

Here are the key characteristics:

- **Data Structure**:
  Data is stored in **tables** (formerly called entities), which contain **rows** (records) and **columns** (fields). You can use **standard tables** for common business scenarios or create **custom tables** for your specific needs. microsoft
- **Security & Compliance**:
  Built on **Microsoft Azure**, Dataverse offers **role-based security**, **row-level and column-level access control**, and compliance with global standards. geeksforgeeks
- **Integration**:
  It connects with **Dynamics 365**, **Microsoft 365 services**, and external systems via connectors (e.g., SAP, Salesforce, Azure SQL). This makes it ideal for unifying data from multiple sources. vantage365

Power Platform products

Use Microsoft Power Platform to build AI-driven agents using Copilot Studio, apps using Power Apps, automate tasks using Power Automate, analyze data using Power BI, and create websites using Power Pages.



Copilot Studio

- Copilot Studio documentation
- Copilot Studio training
- Copilot Studio videos



Power Apps

- Power Apps documentation
- Power Apps training
- Power Apps videos



Power Automate

- Power Automate documentation
- Power Automate training
- Power Automate videos



Power BI

- Power BI documentation
- Power BI training
- Power BI videos



Power Pages

- Power Pages documentation
- Power Pages training
- Power Pages videos

COMMON DATA MODEL SCHEMA

Dynamics 365

CORE

- Account
- Activity
- Contact
- Currency
- Email
- Goal
- Letter
- Note
- Owner
- Organization
- Position
- Task
- ...

- Account
- Appointment
- Campaign
- Contact
- Lead
- Marketing List
- Phone Call
- Social Activity
- ...

SALES

- Competitor
- Discount
- Invoice
- Opportunity
- Order
- Order Product
- Quote
- ...

SERVICE

- Case
- Contract
- Entitlement
- Resource
- Service
- Scheduling Group
- Task
- ...

FINANCE

- Main accounts
- General Ledger
- Budget
- Bank
- Fixed assets
- Dimensions
- ...

SUPPLY CHAIN

- Product Information Management
- Procurement
- Trade Agreements
- Sales
- Production
- Warehouse and Transportation Management
- Asset Management
- ...

COMMERCE

- Channel
- Product catalog
- Prices & Discounts
- Sales transaction
- Payment instrument
- Customer loyalty
- Worker
- ...

MARKETING

- Account
- Contact
- Event
- Marketing Email
- Marketing Page
- ...

Working with subject matter & industry experts, Common Data Model is expanding to include additional business entities and concepts

WEB

- Link Clicks
- Web Interactions
- Web Page
- ...

PROJECT OPERATIONS

- Project Planning
- Project Resourcing
- Project Time and Expense
- Project Invoicing
- Project Accounting
- ...

HUMAN RESOURCES

- Department
- Job Positions
- Worker
- ...

GLOBALIZATION

- Tax determination and calcultaion
- E-invoicing & real time reporting
- Payment formats
- Tax and audit reports
- Business documents
- Fiscal integrations for Retail
- ...

HEALTHCARE

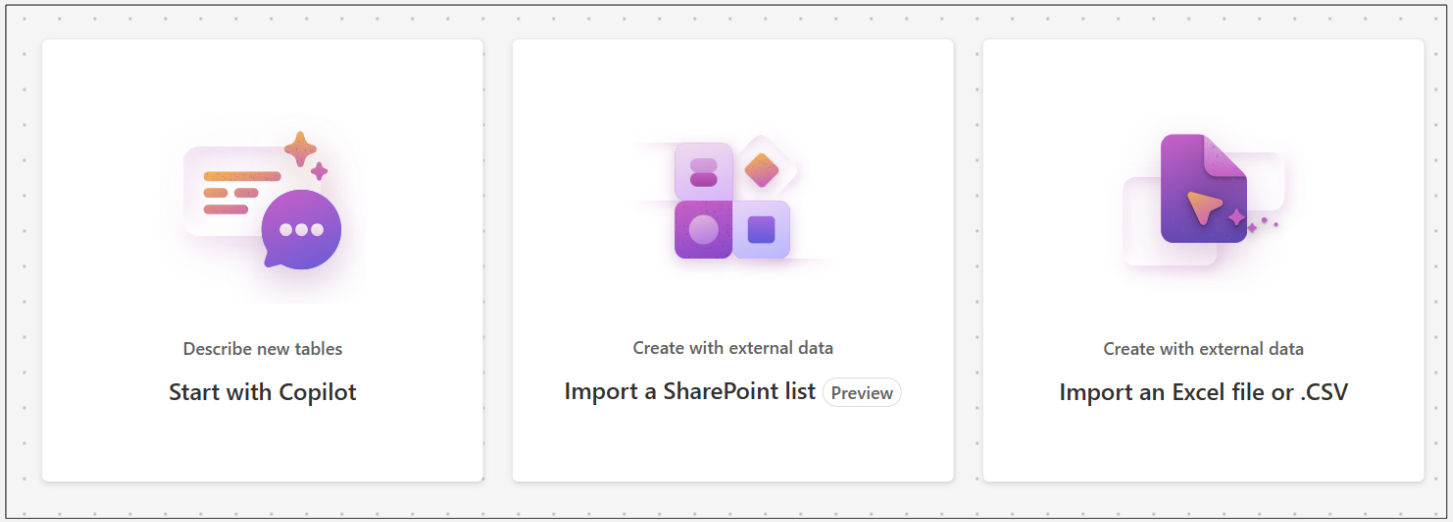
- Patient
- Practicioner
- Device
- Care Plan
- Risk Assessment
- Medication
- Procedure
- Encounter
- Episode of Care
- ...

Account:
Business that represents a customer or potential customer.

Attributes:

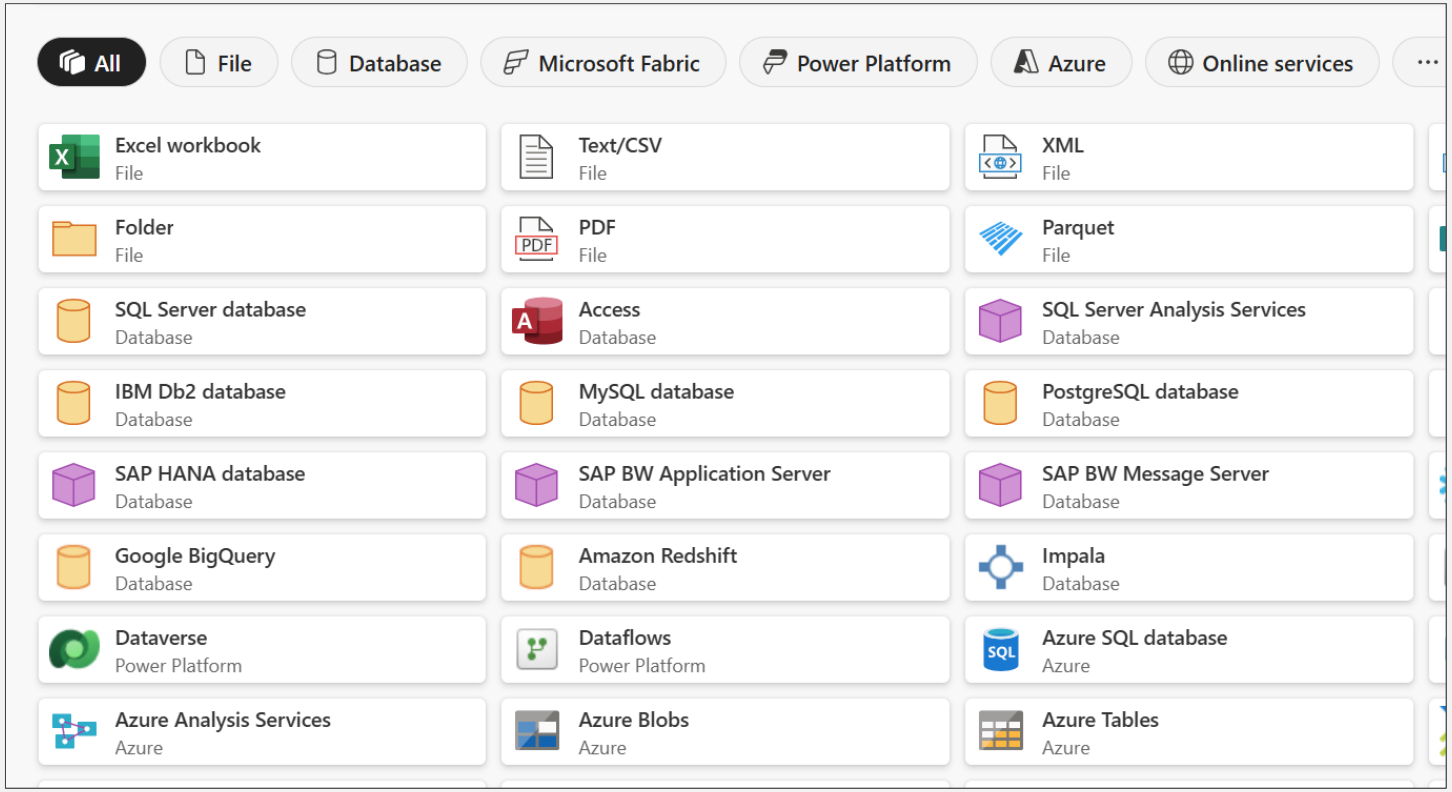
- accountNumber
- accountRatingCode
- createdOn
- creditLimit
- openDeals
- openRevenue
- territoryid
- hotelGroup
- ...

No-Code Users



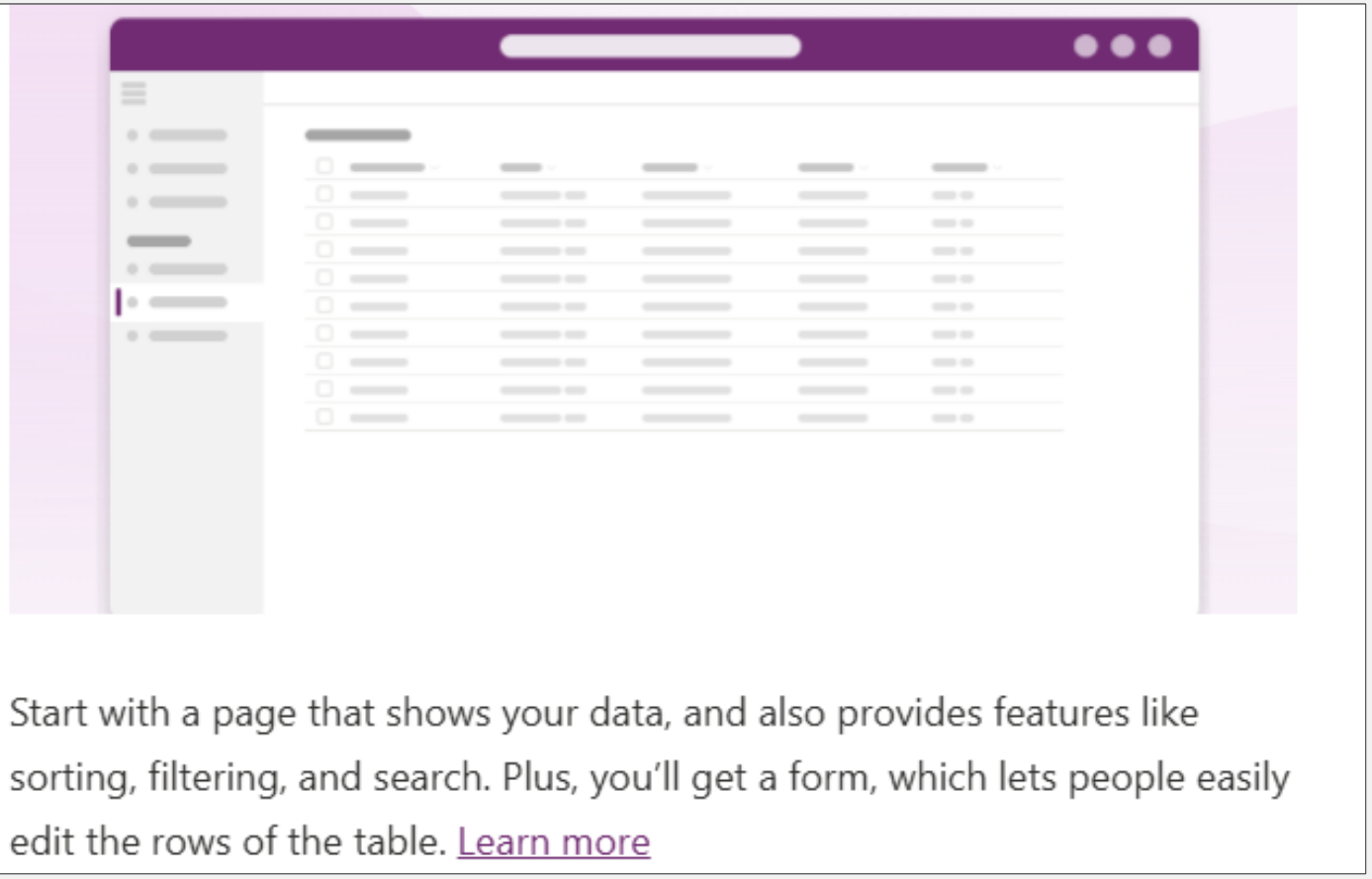
- ❑ CRUD
- ❑ Power Query
- ❑ Model-driven apps

No-Code Users



- ❑ CRUD
- ❑ Power Query
- ❑ Model-driven apps

No-Code Users



Start with a page that shows your data, and also provides features like sorting, filtering, and search. Plus, you'll get a form, which lets people easily edit the rows of the table. Learn more

- ❑ CRUD
- ❑ Power Query
- ❑ Model-driven apps

Low-Code Makers

> **Note**
>
> Delegation warnings help you manage your app so it returns correct results. If the data in your data source exceeds 500 records and a function isn't delegated, Power Fx marks the formula with a blue underline.

- ❑ Power Apps
- ❑ Power Automate
- ❑ Power BI

Low-Code

> **Note**
>
> Delegation warnings help you mana
> results. If the data in your data sour
> function isn't delegated, Power Fx n
> underline.

Operation/Function [1]	Number	Text	Boolean	DateTime	Complex [2]
`<, <=,<>, >, >=`	Yes [3]	No	No	Yes	Yes
`=`	Yes	Yes	Yes	Yes	Yes
`Filter`	Yes	Yes	Yes	Yes	Yes
`IsBlank`	-	No [4]	-	-	No
`Lookup`	Yes	Yes	Yes	Yes	Yes
`Sort`	Yes	Yes	Yes	Yes	No
`SortByColumns`	Yes	Yes	Yes	Yes	No
`StartsWith`	-	Yes	-	-	Yes [5]
`UpdateIf/RemoveIf` [6]	Yes	No	-	-	No

Low-Code

> **Note**
>
> Delegation warnings help you mana
> results. If the data in your data sour
> function isn't delegated, Power Fx n
> underline.

Operation/Function
`<, <=,<>, >, >=`
`=`
`Filter`
`IsBlank`
`Lookup`
`Sort`
`SortByColumns`
`StartsWith`
`UpdateIf/RemoveIf` [6]

Operation or function	Number	Text	Boolean	DateTime	Guid
`*, +, -, /`	Yes	-	-	No	-
`<, <=, >, >=`	Yes	No	No	Yes	-
`=, <>`	Yes	Yes	Yes	Yes	Yes
`Average`	Yes	-	-	-	-
`EndsWith`	-	Yes [1]	-	-	-
`Filter`	Yes	Yes	Yes	Yes [2]	Yes
`In` (substring)	-	Yes [3]	-	-	-
`IsBlank` [4]	No	No	No	No	No
`Len` (length)	-	Yes [5]	-	-	-
`Lookup`	Yes	Yes	Yes	Yes	Yes
`Max`	Yes	-	-	No	-
`Min`	Yes	-	-	No	-
`Search`	No	Yes	No	No	-
`Sort`	Yes	Yes	Yes	Yes	-
`SortByColumns`	Yes	Yes	Yes	Yes	-
`StartsWith`	-	Yes [6]	-	-	-
`Sum`	Yes	-	-	-	-
`UpdateIf, RemoveIf` [7]	Yes	Yes	Yes	Yes	Yes

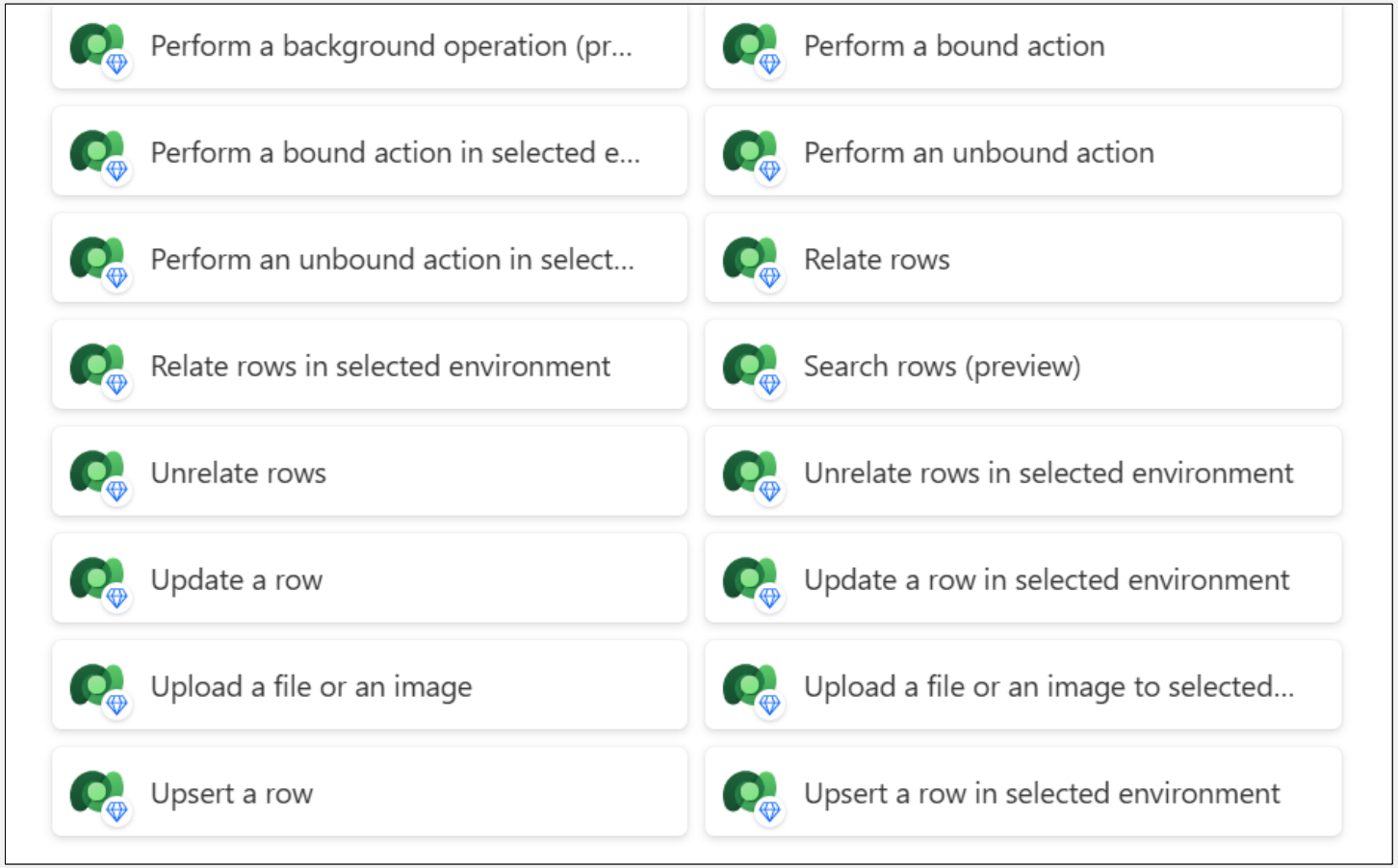
Low-Code

> **Note**
>
> Delegation warnings help you mana
> results. If the data in your data sour
> function isn't delegated, Power Fx n
> underline.

Operation/Fu
<, <=,<>, >,
=
Filter
IsBlank
Lookup
Sort
SortByColumns
StartsWith
UpdateIf/Remo

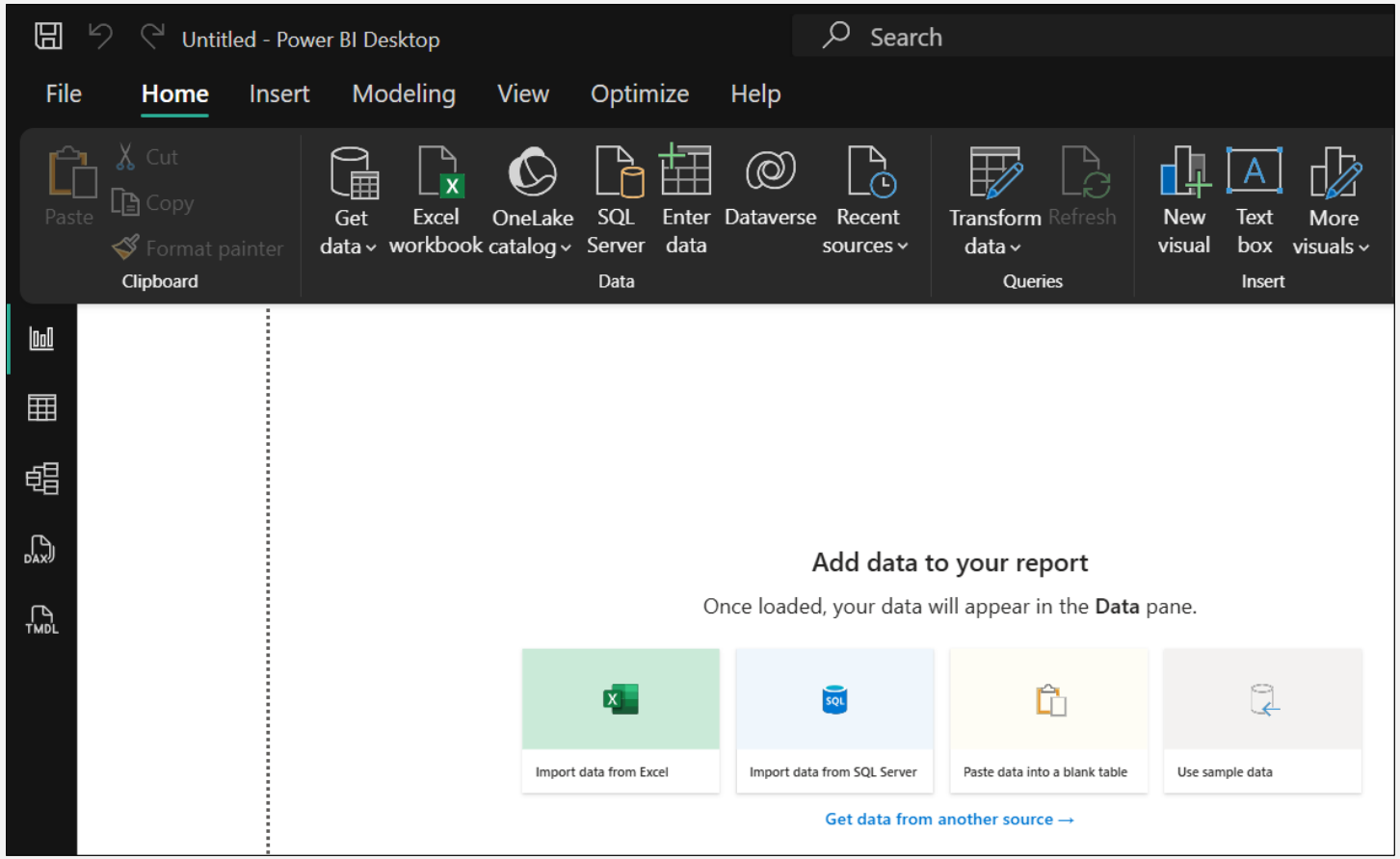
Item	Number [1]	Text [2]	Choice	DateTime [3]	Guid
<, <=, >, >=	Yes	Yes	No	Yes	-
=, <>	Yes	Yes	Yes	Yes	Yes
And/Or/Not	Yes	Yes	Yes	Yes	Yes
CountRows [4] [5], CountIf [6]	Yes	Yes	Yes	Yes	Yes
Filter	Yes	Yes	Yes	Yes	Yes
First [7]	Yes	Yes	Yes	Yes	Yes
In (membership) [8]	Yes	Yes	Yes	Yes	Yes
In (substring)	-	Yes	-	-	-
IsBlank [9]	Yes	Yes	No	Yes	Yes
Lookup	Yes	Yes	Yes	Yes	Yes
Search	No	Yes	No	No	-
Sort	Yes	Yes	Yes	Yes	-
SortByColumns	Yes	Yes	Yes	Yes	-
StartsWith	-	Yes	-	-	-
Sum, Min, Max, Avg [6]	Yes	-	-	No	-
UpdateIf/RemoveIf [10]	Yes	-	-	No	-

Low-Code Makers



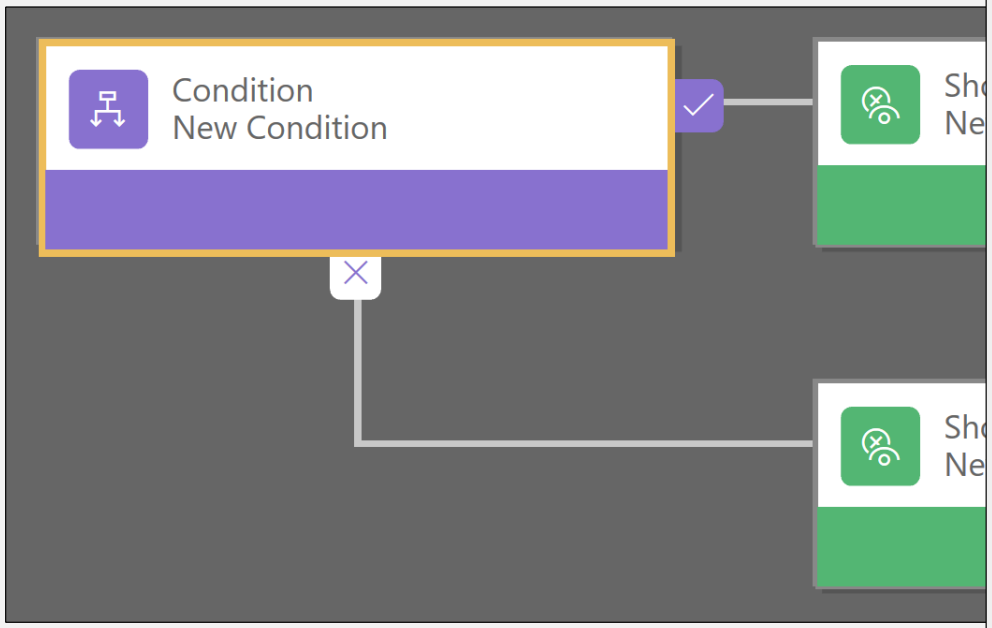
- ❑ Power Apps
- ❑ Power Automate
- ❑ Power BI

Low-Code Makers



- ❑ Power Apps
- ❑ Power Automate
- ❑ Power BI

Pro-Code D



Actions

- Recommendation
- Lock/ Unlock
- Show Error Message
- Set Field Value
- Set Default Value
- Set Business Required
- Set Visibility

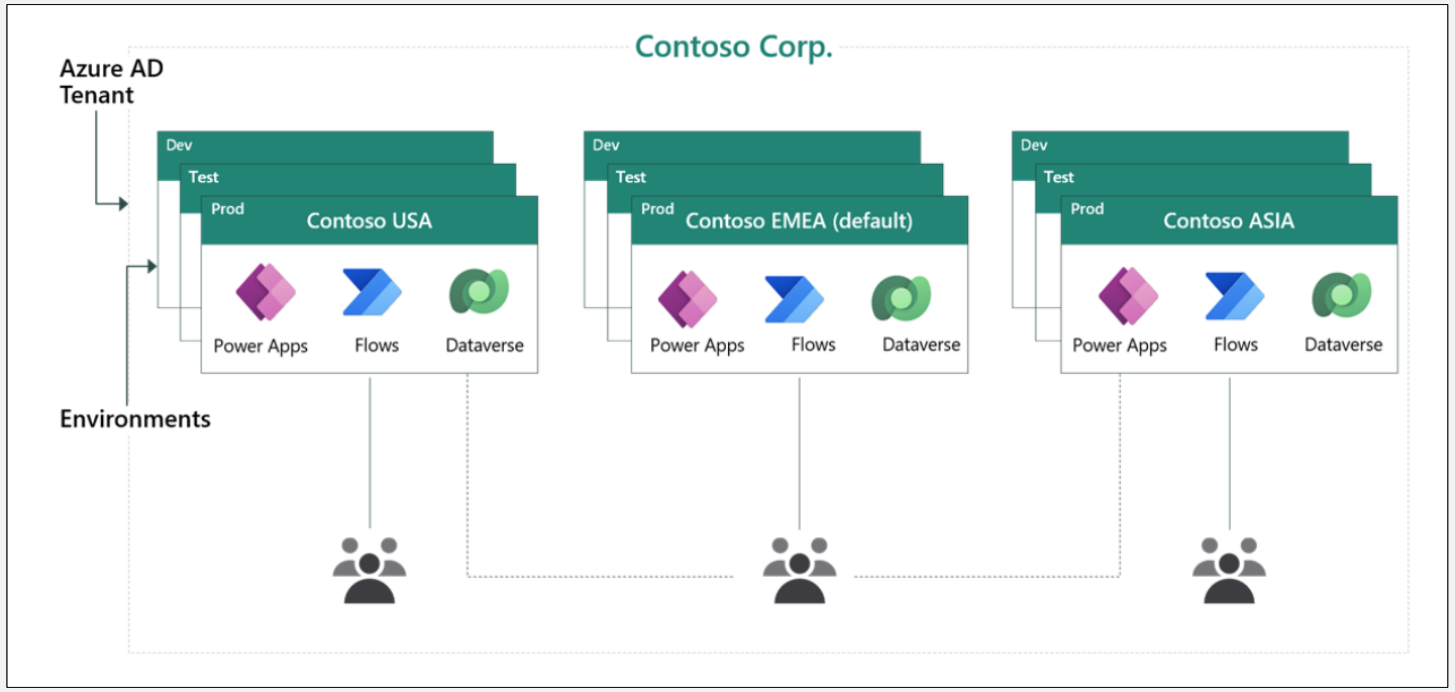
Pro-Code Devs

What is XrmToolBox?

XrmToolBox is a Windows application that connects to Microsoft Dataverse. Dynamics 365 Customer Engagement (CE) applications, like Dynamics 365 for Sales, Service or Talent also use Microsoft Dataverse as their foundational data service. XrmToolBox, provides tools to ease customization, configuration and operation tasks for anything built on Microsoft Dataverse, including Dynamics 365 CE (formerly CRM) and model-driven PowerApps. It is shipped with more than 30 tools to make administration, customization or configuration tasks easier and less time consuming. And more than 100 other tools are available in the Tool Library.

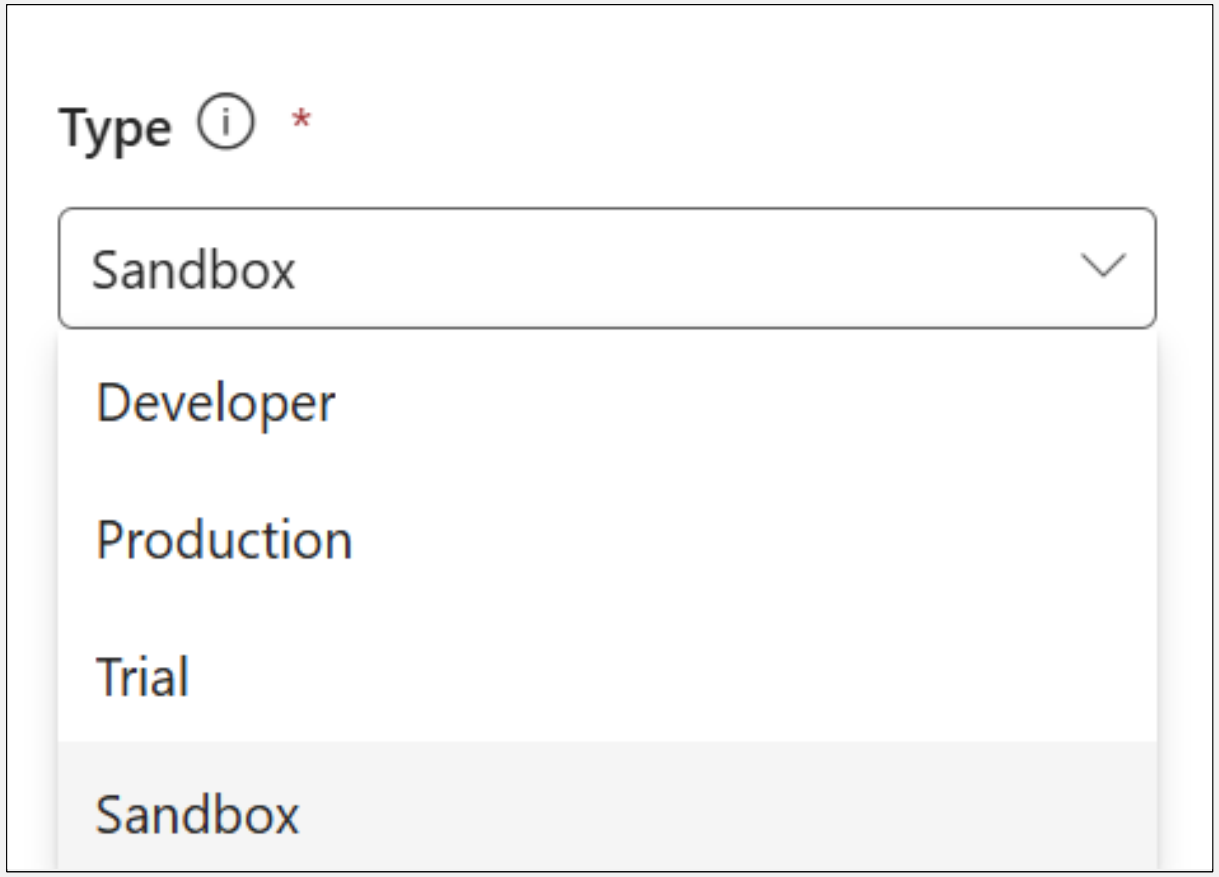
- ❑ Business Rules
- ❑ Custom Code
- ❑ Application Lifecycle Management

Pro-Code Devs



- ❑ Business Rules
- ❑ Custom Code
- ❑ Application Lifecycle Management

Database Engineers



- ❑ Environments
- ❑ Security Roles
- ❑ Geo Regions

Database Engineers

Role ↑		Business unit	Managed
Activity Viewer	···	orgda13f916	Yes
Agent 365 Tools Role	···	orgda13f916	Yes
Agent Viewer	···	orgda13f916	Yes
AIB Roles	···	orgda13f916	Yes
AIB SML Roles	···	orgda13f916	Yes
ApolloServiceRole	···	orgda13f916	Yes
App Deployment Orchestration Role	···	orgda13f916	Yes
App Opener	···	orgda13f916	Yes

- ❑ Environments
- ❑ Security Roles
- ❑ Geo Regions

Database Engineers



- ❑ Environments
- ❑ Security Roles
- ❑ Geo Regions

Demo

- Tables
  - Standard
  - Activity
  - Custom
  - Virtual
- Schema
  - Columns
  - Keys
  - Relationships
- Apps
  - Model-driven
  - Canvas

Recap:



No-Code Users



Low-Code Makers



Pro-Code Devs



Database Engineers

Are you building solutions for the enterprise, but still storing your business data in SQL Server, SharePoint Online, Excel and/ or Access? If so, then you may need to reevaluate and consider Microsoft Dataverse, the centralized, native data solution of the Power Platform.

For many of your internal low-code and pro-code initiatives, Dataverse is likely the better data storage offering. Designed to be a secure, low-code solution, Microsoft built Dataverse to help organizations increase their return on investment (ROI), better scale their in-house applications, standardize their data sources, and promote rapid innovation, with the help of Copilot.

Let's dive into Microsoft Dataverse, explore jumpstarting your flex team's productivity, and tackling your project backlog.

Low-Code Makers

Database Engineers

Contact:

/The2Lakes

@The2Lakes

/losodamus



Charles is a technical speaker and low-code advocate with a pro-code background, specializing in transforming business data architecture. With a passion for innovation, he shares his expertise to solve complex business problems and guide organizations in reimagining their legacy processes, advancing automation initiatives, streamlining operations, and reducing infrastructure costs.